AE4868 – Numerical Astrodynamics

Mission Design - Variational Equations

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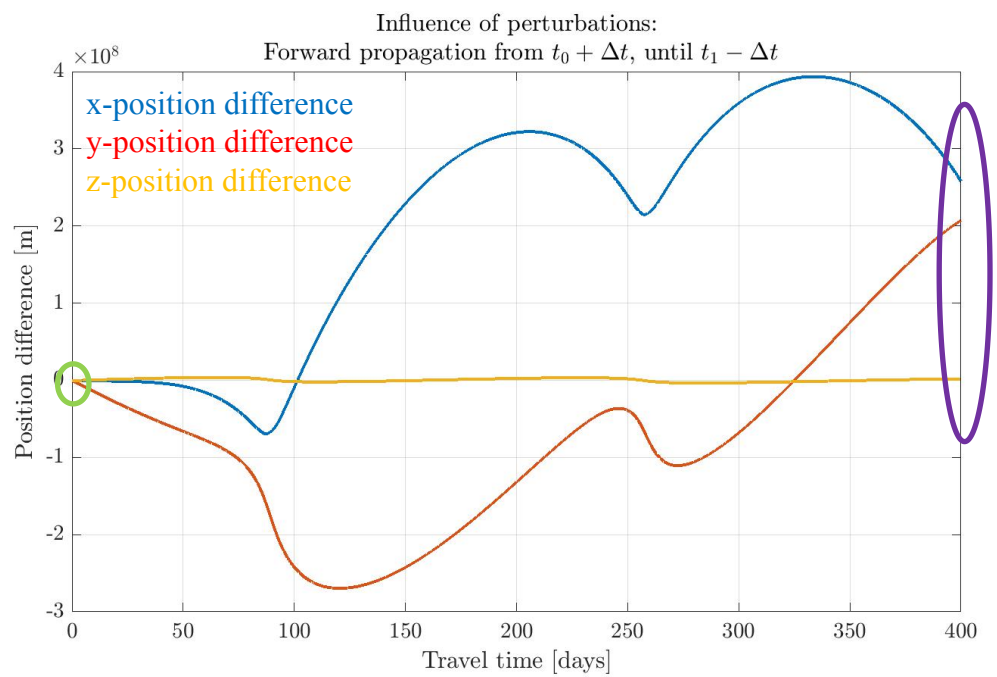
Variational Equations

Introduction

- ***What we have (Lambert problem):***
  - A transfer that works perfectly under specific assumption
  - A transfer that *almost* works for a high-accuracy model

- ***What we have (Numerical model):***
  - A numerical model using Lambert arc as initial state
  - Same initial state: (slightly) different final state



Variational Equations

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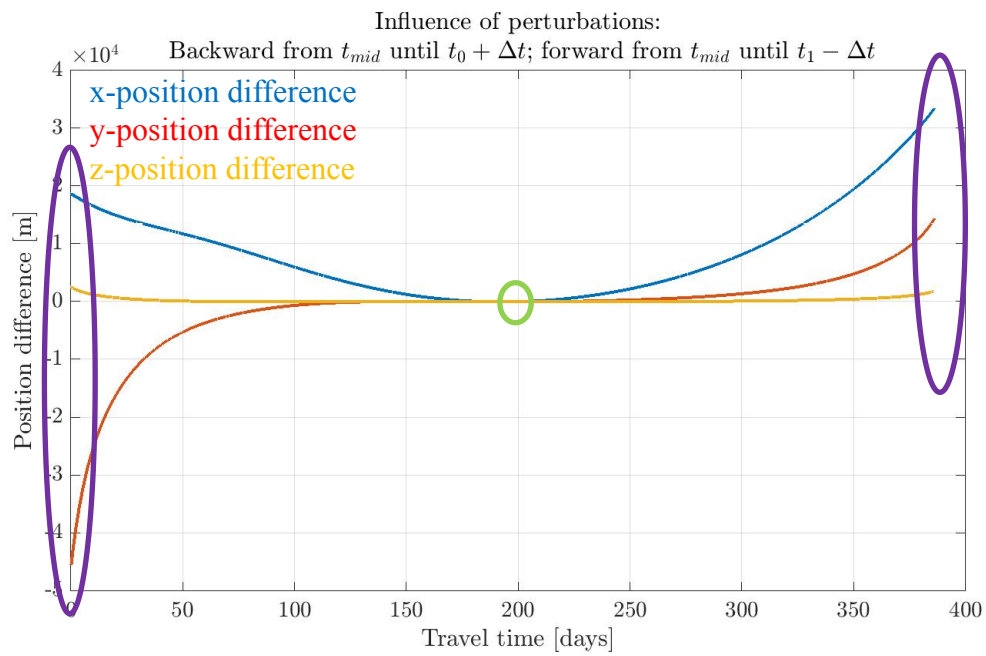

Variational Equations

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- ***What we need:***
  - A tool to correct the numerical model to meet boundary conditions - (depart from Earth, arrive at Mars)

Variational Equations

Introduction

- ***What we need*** - A tool to correct the numerical model

- ***This video:***

  - **Definition** - state transition matrix
  - **Definition** - sensitivity matrix
  - **Calculating** state transition and sensitivity - the variational equations
  - **Solving** variational equations

Variational Equations

Introduction

- ***What we need*** - A tool to correct the numerical model
  - Making modifications to our orbit design so that our numerical trajectory meets all constraints (arrival and departure)
  - Which modifications can we make?

- We can change
  - Initial state
  - Acceleration model

$$\dot{\mathbf{x}} = \boxed{\mathbf{f}\left(\mathbf{x}, t; \mathbf{p}\right)}$$

$$\boxed{\mathbf{x}(t_0) = \mathbf{x}_0}$$

- Changing the acceleration model
  - We want our numerical model to be **high-accuracy**
  - We have **no control** over our physical environment
  - But, we can choose to **use and control thrust**!

Variational Equations

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$$\dot{\mathbf{x}} = \boxed{\mathbf{f}\left(\mathbf{x}, t; \mathbf{p}\right)}$$

$$\boxed{\mathbf{x}(t_0) = \mathbf{x}_0}$$

- We'll introduce two tools
  - **State transition matrix** change in numerical solution due to change in $\mathbf{x}_0$
  - **Sensitivity matrix $\mathbf{S}$** change in numerical solution due to change in dynamical model

Variational Equations

Definition - state transition matrix

- State transition matrix
  - By definition: linearized change in state $\mathbf{x}$ at time t, due to change in initial state $\mathbf{x}_0$:

$$\mathbf{\Phi}(t, t_0) = \frac{\boxed{\partial \mathbf{x}(t)}}{\boxed{\partial \mathbf{x}_0}}$$

  - What does it mean
    - If our initial state $\mathbf{x}_0(=\mathbf{x}(t_0))$ changes: how will our later state $\mathbf{x}(t)$ change?
    - **Change in initial state changes the solution to the equations of motion**

Variational Equations

Definition - sensitivity matrix

- Sensitivity $\mathbf{S}$
  - By definition: linearized change in state $\mathbf{x}$ at time t, due to change in some dynamical model parameter(s) $\mathbf{p}$

$$\mathbf{S}(t) = \frac{\boxed{\partial \mathbf{x}}}{\boxed{\partial \mathbf{p}}}$$

  - What does it mean
    - If we change parameters in the equations of motion, how does our states change as a function of time?
    - **Change in dynamical model parameter changes the solution to the equations of motion**

Variational Equations

Definition - sensitivity matrix

- Sensitivity $\mathbf{S}$
  - By definition: linearized change in state $\mathbf{x}$ at time t, due to change in some dynamical model parameter(s) $\mathbf{p}$

$$\mathbf{S}(t) = \frac{\boxed{\partial \mathbf{x}}}{\boxed{\partial \mathbf{p}}}$$

  - Examples for 'entries' of $\mathbf{p}$
    - Gravitational parameter
    - Spherical harmonic coefficients
    - Drag coefficient
    - Thrust parameters
    - ...

Variational Equations

Definition - summary

- State transition matrix linearized change in state $\mathbf{x}$ at time t, due to change in initial state $\mathbf{x}_0$:

$$\mathbf{\Phi}(t, t_0) = \frac{\partial \mathbf{x}(t)}{\partial \mathbf{x}_0} \longrightarrow \boxed{\Delta \mathbf{x}(t) \approx \Phi(t, t_0) \Delta \mathbf{x}_0}$$

- Sensitivity $\mathbf{S}$ linearized change in state $\mathbf{x}$ at time t, due to change in some dynamical model parameter(s) $\mathbf{p}$

$$\mathbf{S}(t) = \frac{\partial \mathbf{x}}{\partial \mathbf{p}} \longrightarrow \boxed{\Delta \mathbf{x}(t) \approx \mathbf{S}(t) \Delta \mathbf{p}}$$

- **Models assume linearity**
  - Only work for 'small' changes in $\mathbf{x}_0$ and $\mathbf{p}$

Variational Equations

Calculation - state transition matrix

- The state transition matrix is computed from:

$$\frac{d}{dt}\left(\Phi(t,t_0)\right) = \left(\frac{\partial \dot{\mathbf{x}}}{\partial \mathbf{x}}\right)\Phi(t,t_0)$$

$$\Phi(t_0,t_0) = \mathbf{1}_{6\times 6}$$

Variational Equations

Calculation - state transition matrix

- The state transition matrix is computed from:

$$\boxed{\frac{d}{dt}\left(\frac{\partial \mathbf{x}}{\partial \mathbf{x}_0}\right)} = \boxed{\left(\frac{\partial \dot{\mathbf{x}}}{\partial \mathbf{x}}\right)}\boxed{\frac{\partial \mathbf{x}}{\partial \mathbf{x}_0}}$$

$$\frac{\partial \mathbf{x}_0}{\partial \mathbf{x}_0} = \mathbf{1}_{6\times 6}$$

- What this equation means:
  - A change in initial state will cause a change in later state
  - If we are at a different state, our accelerations will be different
  - A change in accelerations will cause a further difference in state

Variational Equations

Calculation - sensitivity matrix

- The sensitivity matrix is computed from:

$$\frac{d}{dt}\left(\mathbf{S}(t)\right) = \left(\frac{\partial \dot{\mathbf{x}}}{\partial \mathbf{x}}\right)\mathbf{S}(t) + \left(\frac{\partial \dot{\mathbf{x}}}{\partial \mathbf{p}}\right)$$

$$\mathbf{S}(t_0) = \mathbf{0}_{N_p \times 6}$$

- What this equation means:
  - A change in **parameters** will cause a change in later state
  - If we are at a different state, our accelerations will be different
  - A change in accelerations will cause a further difference in state
  - If our **parameters** our different, our acceleration will be different

Variational Equations

Calculation

- ***We now have***
  - Differential equations governing state transition and sensitivity matrices - **the variational equations**

$$\frac{d}{dt}\left(\Phi(t,t_0)\right) = \boxed{\left(\frac{\partial \dot{\mathbf{x}}}{\partial \mathbf{x}}\right)}\Phi(t,t_0)$$

$$\Phi(t_0,t_0) = \mathbf{1}_{6\times 6}$$

$$\frac{d}{dt}\left(\mathbf{S}(t)\right) = \boxed{\left(\frac{\partial \dot{\mathbf{x}}}{\partial \mathbf{x}}\right)}\mathbf{S}(t) + \boxed{\left(\frac{\partial \dot{\mathbf{x}}}{\partial \mathbf{p}}\right)}$$

$$\mathbf{S}(t_0) = \mathbf{0}_{N_p\times 6}$$

  - To evaluate, we need **partial derivatives of accelerations**, w.r.t
    - Current state
    - Parameter values

Variational Equations

Calculation - partial derivatives

- Computing the partial derivates - an easy example $\mathbf{a} = -\frac{\mu}{r^3}\mathbf{r}$

$$\frac{\partial \mathbf{a}}{\partial \mathbf{r}} = -\mu\left(\boxed{\frac{\mathbf{1}_{3\times 3}}{r^3}} - \boxed{\frac{3\mathbf{r}\cdot\mathbf{r}^T}{r^5}}\right)$$

$$\frac{\partial \mathbf{a}}{\partial \mathbf{v}} = \boxed{\mathbf{0}_{3\times 3}}$$

$$\frac{\partial \mathbf{a}}{\partial \mathbf{p}} = \boxed{-\frac{1}{r^3}\mathbf{r} = \frac{\mathbf{a}}{\mu}}$$

(for the case $\mathbf{p}$=[ ])

- The dependency on position is twofold!
- No dependency on velocity
- The dependency on gravitational parameter is very straightforward

Variational Equations

Calculation - partial derivatives

- Computing the partial derivates - some are a lot harder!

$$U_B(\mathbf{r}_{BA}) = \frac{\mu}{r} \sum_{l=0}^{l_{max}} \sum_{m=0}^{l} \left(\frac{R}{r}\right)^l \bar{P}_{lm}(\sin\phi) \left(\bar{C}_{lm} \cos m\theta + \bar{S}_{lm} \sin m\theta\right)$$

$$\mathbf{a}_{BA} = \nabla U_B(\mathbf{r}_{BA})$$

- An analytical formulation for the partial derivatives exists, but is lengthy

Variational Equations

Calculation - partial derivatives

- Computing the partial derivates - some seem easy, but are a nightmare!

$$\mathbf{a}_{BA} = -\left(\frac{\rho v_{\text{air}}^2 C_D S_{ref}}{2m}\right)\hat{\mathbf{v}}_{\text{air}}$$

- Density
  - May or may not depend on (spherical?) position
- Drag coefficient
  - May or may not depend on position and velocity
- Airspeed:
  - Clear dependence on velocity
  - May or may not depend on position (and have additional velocity dependence!) through model of relative wind

Variational Equations

Generating a solution

- Calculating **full** state transition and sensitivity matrices is cumbersome
  - All relevant routines have been implemented in Tudat
  - Define your problem
  - Then: just push the button!

- But, what goes on 'behind the scenes'?

Variational Equations

Generating a solution

- Let's put it all together - we want to propagate
  - State transition matrix
  - Sensitivity matrix **S**
  - State vector **x**

Variational Equations

Generating a solution

- Let's put it all together - we want to propagate
  - State transition matrix
  - Sensitivity matrix **S**
  - State vector **x**

- These are propagated **together** - we introduce

$$\mathbf{Y} = \left[\Phi \vdots \mathbf{S} \vdots \mathbf{x} \right]$$

- And we now need to define our differential equation:

$$\dot{\mathbf{Y}} = \mathbf{F}(\mathbf{Y}, t)$$

- For which we have all the ingredients!

Variational Equations

Generating a solution - summary

- Matrix differential equation for $\mathbf{Y}$

$$\mathbf{Y} = \left[\Phi \vdots \mathbf{S} \vdots \mathbf{x} \right]$$

$$\dot{\mathbf{Y}} = \mathbf{F}(\mathbf{Y}, t)$$

$$\mathbf{F}(\mathbf{Y}, t) = \frac{\partial \dot{\mathbf{x}}}{\partial \mathbf{x}} \cdot \left[\Phi \vdots \mathbf{S} \vdots \mathbf{0} \right] + \left[\mathbf{0} \vdots \frac{\partial \dot{\mathbf{x}}}{\partial \mathbf{p}} \vdots \mathbf{0} \right] + \left[\mathbf{0} \vdots \mathbf{0} \vdots \dot{\mathbf{x}} \right]$$

- Stick it into a numerical integrator and you get your results!
  - State transition matrix
  - Sensitivity matrix **S**
  - State vector **x**

- Why is this useful? Many reasons!
  - One application: differential correction